Financial Computing **National Taiwan University**

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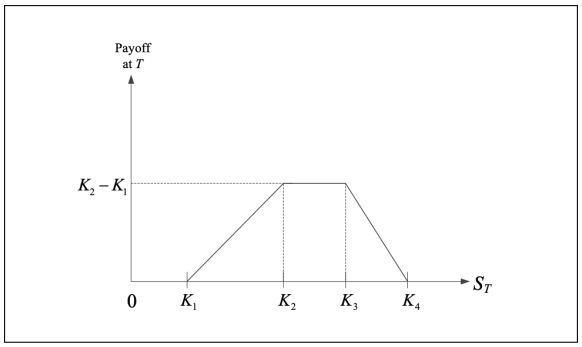
Prerequisite

- $\frac{dS}{S} = \mu dt + \sigma dZ$
- $lnS_T = lnS_0 + (r - q - \frac{\sigma^2}{2})T + \sigma \Delta Z^Q(T)$
- $N(\cdot)$ denotes the CDF of the standard normal distribution.

Basic Requirement



i) **Derivation of the closed-form formula:** Let $f(S_T)$ be the payoff function. Under the risk-neutral measure $\mathbb{Q}$, the asset price S_T follows a lognormal distribution, and the present value of the option is discounted expected payoff:

$$V_0 = e^{-rT} E^{\mathbb{Q}}[Payoff(S_T)]$$

The payoff can be decomposed into three indicator functions 1_A, 1_B, and 1_C:

$$E^Q[Payoff(S_T)] = E^Q[(S_T - K_1)1_A + (K_2 - K_1)1_B + ((K_2 - K_1) - m(S_T - K_3))1_C]$$

where $m = \frac{K_2 - K_1}{K_4 - K_3}$ and the events are:

- $A = \{K_1 \leq S_T < K_2\}$
- $B = \{K_2 \leq S_T < K_3\}$
- $C = \{K_3 \leq S_T < K_4\}$

For $[K1, K2]$, $E^Q[(S_T - K_1)1_A] = E^Q[S_T \cdot 1_A] - K_1 E^Q[1_A]$

- The option is in-the-money if $S_T > K$
  1. $S_0 e^{(r-q-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}Z} > K$
  2. $(r - q - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z > \ln\left(\frac{K}{S_0}\right)$
  3. $\sigma\sqrt{T}Z > \ln\left(\frac{K}{S_0}\right) - (r - q - \frac{1}{2}\sigma^2)T$
  4. $\sigma\sqrt{T}Z > -\ln\left(\frac{S_0}{K}\right) - (r - q - \frac{1}{2}\sigma^2)T$
  5. $Z > \frac{-\ln(S_0/K)-(r-q-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$
  6. $Z < \frac{\ln(S_0/K)+(r-q-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$, By the symmetry of normal distribution
  7. Define $d_2(K) = \frac{\ln(S_0/K)+(r-q-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$

$d_2(K)$ is the z-score of S_T reaching K, and $N(d_2(K))$ is the probability of S_T reaching K. Therefore:

$$E^Q[1_A] = N(d2(K1)) - N(d2(K2))$$
$$E^Q[1_B] = N(d2(K2)) - N(d2(K3))$$
$$E^Q[1_C] = N(d2(K3)) - N(d2(K4))$$

To calculate $E^Q[S_T \cdot 1_A]$, we define a Radon-Nikodym derivative $\Lambda = e^{-\frac{\sigma^2}{2}T+\sigma\Delta Z^Q(T)}$ to change from measure Q to measure R. Setting $H(t) = -\sigma$ in the Girsanov theorem, the relation between Q and R measure would be:

$$dZ^R = dZ^Q - \sigma dt$$

The drift term in R measure would change to:

$$lnS_T = lnS_0 + (r - q + \frac{\sigma^2}{2})T + \sigma\Delta Z^R(T)$$

$$d_1(K) = \frac{\ln(S_0/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

Applying $E^Q[\Lambda \cdot 1_A] = E^R[1_A]$:

$$E^Q[S_T \cdot 1_A] = S_0 e^{(r-q)T} N(d_1(K_1)) - S_0 e^{(r-q)T} N(d_1(K_2))$$

For $[K2, K3]$, the payoff is a constant $K_2 - K_1$. We only need the risk-neutral probability:

$$E^Q[(K_2 - K_1)1_B] = (K_2 - K_1)[N(d_2(K_2)) - N(d_2(K_3))]$$

For $[K3, K4]$, the slope is $m = \frac{K_2 - K_1}{K4 - K3}$ and the payoff function is $m(K_4 - S_T)$

$$E^Q[m(K_4 - S_T)1_C] = mK_4E^Q[1_C] - mE^Q[S_T 1_C]$$

$$E^Q[m(K_4 - S_T)1_C] = mK_4E^Q[1_C] - mE^Q[S_T 1_C]$$

Similar to previous $Q \to R$:

$$E^Q[m(K_4 - S_T)1_C] = mK_4[N(d_2(K_3)) - N(d_2(K_4))] - mS_0e^{(r-q)T}[N(d_1(K_3)) - N(d_1(K_4))]$$

Finally, add everything and discount by e^{-rT}:

$$V_0 = S_0e^{-qT}[N(d_1(K_1)) - N(d_1(K_2))] - K_1e^{-rT}[N(d_2(K_1)) - N(d_2(K_2))]$$

$$+(K_2 - K_1)e^{-rT}[N(d_2(K_2)) - N(d_2(K_3))]$$

$$+mK_4e^{-rT}[N(d_2(K_3)) - N(d_2(K_4))] - mS_0e^{-qT}[N(d_1(K_3)) - N(d_1(K_4))]$$

Where $N(\cdot)$, $d_1(K)$, $d_2(K)$, and m are:

$$N(x) = ND(0,1).cdf(x)$$

$$d_1(K) = \frac{\ln(S_0/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$$d_2(K) = \frac{\ln(S_0/K) + (r - q - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$$m = \frac{K_2 - K_1}{K_4 - K_3}$$

ii) The pseudo code of the program is as follow, the detail python script can be found in supplementary.

```
def N(x: float) -> float:
    return NormalDist(0.0, 1.0).cdf(x)

def calc_d1(S0, K, T, r, q, sigma)
    return (
        (math.log(S0 / K) + (r - q + 0.5 * sigma**2) * T)
        / (sigma * math.sqrt(T)
    )

def calc_d2(d1_val, T, sigma):
    return d1_val - sigma * math.sqrt(T)

# d_K = calc_d(S0, K, T, r, q, sigma)

part1_asset = S0 * math.exp(-q * T) * (N(d1_K1) - N(d1_K2))
part1_cash  = K1 * math.exp(-r * T) * (N(d2_K1) - N(d2_K2))
segment_1   = part1_asset - part1_cash

```

```
segment_2 = (K2 - K1) * math.exp(-r * T) * (N(d2_K2) - N(d2_K3))

part3_cash  = m * K4 * math.exp(-r * T) * (N(d2_K3) - N(d2_K4))
part3_asset = m * S0 * math.exp(-q * T) * (N(d1_K3) - N(d1_K4))
segment_3   = part3_cash - part3_asset

V0 = segment_1 + segment_2 + segment_3
```

Bonus 1

By reducing the derivation above, we can actually get:

- Strike K_1 terms: $S_0e^{-qT}N(d_1(K_1)) - K_1e^{-rT}N(d_2(K_1)) = \mathbf{C(K_1)}$
- Strike K_2 terms: $-S_0e^{-qT}N(d_1(K_2))+K_1e^{-rT}N(d_2(K_2))+(K_2-K_1)e^{-rT}N(d_2(K_2))$
  $\Rightarrow -S_0e^{-qT}N(d_1(K_2)) + K_2e^{-rT}N(d_2(K_2)) = -\mathbf{C(K_2)}$
- Strike K_3 terms: $-(K_2 - K_1)e^{-rT}N(d_2(K_3)) + (K_2 - K_1 + mK_3)e^{-rT}N(d_2(K_3)) - mS_0e^{-qT}N(d_1(K_3))$
  $\Rightarrow mK_3e^{-rT}N(d_2(K_3)) - mS_0e^{-qT}N(d_1(K_3)) = -\mathbf{m} \cdot \mathbf{C(K_3)}$
- Strike K_4 terms: $-(K_2-K_1+mK_4-mK_4+mK_3)e^{-rT}N(d_2(K_4))+mS_0e^{-qT}N(d_1(K_4))$
  $\Rightarrow mS_0e^{-qT}N(d_1(K_4)) - mK_4e^{-rT}N(d_2(K_4)) = \mathbf{m} \cdot \mathbf{C(K_4)}$

```
# Full script can be found in supplementary
m = (K2 - K1) / (K4 - K3)

c1 = bs_call_price(S0, K1, r, q, sigma, T)
c2 = bs_call_price(S0, K2, r, q, sigma, T)
c3 = bs_call_price(S0, K3, r, q, sigma, T)
c4 = bs_call_price(S0, K4, r, q, sigma, T)

total = c1 - c2 - m * c3 + m * c4
```

Bonus 2

The corresponded script can be found in supplementary.

```
Prices = []
Iterate 20 times:
    Price = SimulatePrice(average 10,000 times)
    Prices.append(Price)

```

```
Mean, Std = mean(Prices), std(Prices)
Lowerbound, Upperbound = Mean - 2 * Std, Mean + 2 * Std
return Mean, Lowerbound, Upperbound
```